



Application of Power Electronic for Renewable Generation Wind generation

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DYNAMICS OF RENEWABLES-BASED ENERGY SYSTEMS
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1 Introduction

This report focuses on modeling and analyzing a wind energy generation system using a Permanent Magnet Synchronous Generator (PMSG) and a Voltage Source Converter (VSC) in a back-to-back configuration, forming a Type 4 wind turbine setup. The primary goal is to understand the dynamic behavior and control strategies associated with each subsystem of the model. The modeling and simulation work is carried out using MATLAB and Simulink. The investigation is divided into three core parts. First, a simplified wind turbine model is implemented and tested under various mechanical load scenarios to evaluate its power extraction performance. Next, the PMSG operation is analyzed, with emphasis on the interaction between electrical and mechanical dynamics under VSC control. Finally, a complete Type 4 wind turbine system is assembled by integrating the wind turbine, PMSG, and VSC subsystems. This combined model is then tested under varying wind conditions to assess system response and power output characteristics.

2 General Control Overview

Modern wind energy systems rely on advanced control and conversion technologies to optimize power generation and ensure stable grid integration. This project focuses on a Type 4 wind turbine configuration, where a PMSG is fully decoupled from the grid via a back-to-back VSC. This setup allows full-speed variable operation and independent control of active and reactive power.

The system integrates aerodynamic, mechanical, and electrical subsystems. Wind energy is converted into mechanical torque through the turbine and gearbox, then transformed into electrical power by the PMSG. The VSCs enable simultaneous control of the generator's operation and the connection to the grid. Figure 2 shows the control block diagram for a type 4 wind turbine. Both VSCs use one of their controllable parameters to regulate reactive power. The VSC on the machine side represents the impedance seen by the stator windings, and can thus be used in conjunction with knowledge of the turbine characteristics to maximize power at a given wind speed by controlling the torque load on the PMSG. The VSC on the grid side maintains the DC bus since its active power transfer is determined by the wind and the other VSC.

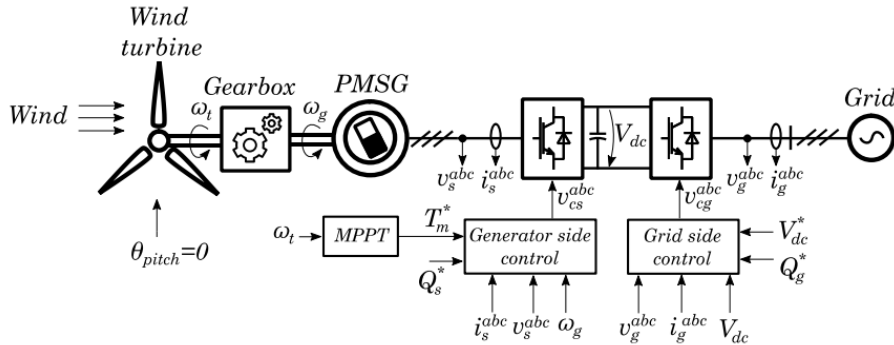


Figure 2: Generation: General Control Overview

The system parameters and control layout, as illustrated in Figure 2 and following tables, serve as the foundation for all simulations and performance analyses in the following tasks.

Parameter	Value
P_n	5 MW
$v_{w,n}$	14 m/s
D_t	88 m
N	80
J_{tot}	$9 \times 10^4 \text{ kg}\cdot\text{m}^2$

(a) Wind Turbine Parameters

Parameter	Value
c_1	0.44
c_2	125
c_3	0
c_4	0
c_5	0
c_6	6.94
c_7	16.5
c_8	0
c_9	-0.0002

(b) Power Coefficient Parameters

Parameter	Value
w_n	$2\pi \cdot 50 \text{ rad/s}$
p	2
R_s	0.01Ω
L_d	0.159 mH
L_q	0.159 mH

(c) PMSG Parameters

Parameter	Value
$S_{c,n}$	5 MVA
$U_{c,n}$	690 V
$V_{dc,n}$	1200 V
L_c	$30.31 \mu\text{H}$ (0.1 pu)
R_c	$0.95 \text{ m}\Omega$ (0.01 pu)
C_{dc}	278 mF (0.04 s)

(d) VSC Back-to-Back Converter

Parameter	Value
U_g	690 V
f_g	50 Hz

(e) AC Grid Parameters

Table 1: Overview of Wind Power System Parameters

3 Wind Turbine Model (Task 3.1)

In the first part of the report the mechanical portion of a wind turbine is modeled in Matlab and Simulink (see Figure 3), to analyze its dynamic response under different mechanical loads.

3.1 Theoretical Overview

The turbine is modeled as a one-mass mechanical system where power extracted from the wind is converted into mechanical torque, which is then applied to various expressions for the generator-side load torque. Shown in Figure 3, the system receives a wind speed input and uses the parameters and characteristic equations shown below to determine the power extracted by the turbine for the given wind speed. The power extracted from the wind is calculated as:

$$P_{wt} = \frac{1}{2} \rho A v_w^3 C_p \quad (1)$$

where $C_p(\Lambda)$ is the power coefficient. The power coefficient at a pitch angle of 0° (which is the case throughout the simulation) is the following:

$$C_p(\Lambda) = c_1 \left(\frac{c_2}{\Lambda} - c_6 \right) e^{-\frac{c_7}{\Lambda}} \quad (2)$$

where

$$\frac{1}{\Lambda} = \frac{1}{\lambda} - c_9 \quad (3)$$

The mechanical torque applied to the shaft is determined in terms of the extracted power:

$$T_m = \frac{P_{wt}}{w_t} \quad (4)$$

The turbine inertia shown in Table 1a is used to calculate the speed of the turbine rotor. Since the rotor speed is used in both sides of the simulation to calculate itself, it is initialized to a small non-zero value. At the generator side, three different mechanical loads are tested. The linear load with load 1: $T_{load,g} = K_1 w_g$, $K_1 = 20$, a quadratic load 2: $T_{load,g} = K_2 w_g^2$, $K_2 = 1$ and an optimal load derived from the maximum power point tracking principle, expressed as:

$$T_{max,g} = \frac{K_{cp}}{N} w_t^2 \quad (5)$$

where the constant K_{Cp} is given as:

$$K_{cp} = \frac{1}{2} \rho A R^3 \frac{c_1 (c_2 + c_6 c_7)^3 e^{\frac{c_2 + c_6 c_7}{-c_2}}}{c_2^2 c_7^4} \quad (6)$$

These load configurations are integrated in the Simulink model to show how the rotor speed, torque, and power coefficient respond to different mechanical loading behaviors.

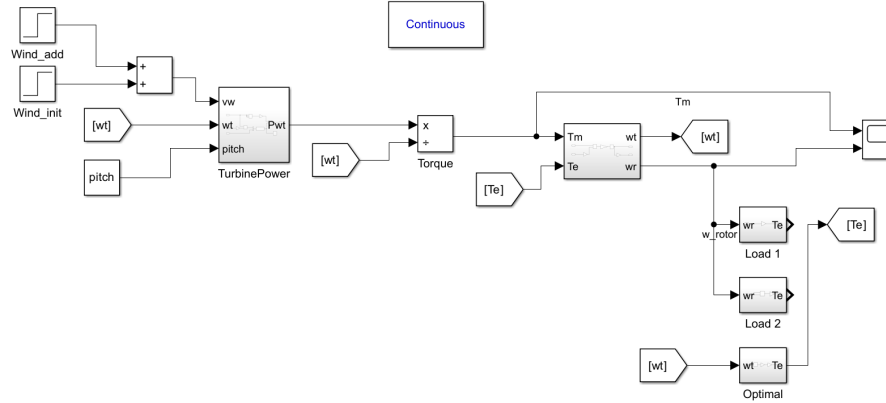


Figure 3: Wind Turbine Model

3.2 Simulation Analysis

The implemented wind turbine model was simulated over a five second interval with a step change in wind speed from $0 - 7\text{m/s}$ at $t = 0\text{s}$ to $7 - 12\text{m/s}$ at $t = 2.5\text{s}$. For each of the three mechanical loads (linear, quadratic and optimal) the rotor speed, generator torque and power coefficient were plotted and analyzed.

3.2.1 Power Coefficient Plots

The power coefficient graph for the three mechanical load types show distinct behaviors in aerodynamic efficiency over time. Load 1 applies a simple constant proportional torque based on the rotor speed. As seen in the graph 4 C_p initially rises but quickly drops and stabilizes at a much lower level. The Load 2 (see 5), which applies a quadratic torque-speed relationship, maintains consistent C_p , with only a small dip at the wind step. These results are expected because the power available to the system increases cubically with wind speed (Equation 1). Therefore, in order for the power coefficient to remain constant, the power extracted by the system must also increase cubically. We know from Eq. 4 that the torque is multiplied with the rotor speed to obtain power. In this case, the rotor speed multiplied with the load torque is the power extracted by the system. Therefore, Load 1's linear torque means that the extracted power is quadratically related to the speed, whereas Load 2's is cubically related. This means that Load 2 presents enough of an increasing load on the generator to extract the same proportion of power regardless of wind speed. The power coefficient is maintained in steady state, but is still lower than the optimal. Load 3 (see 6), based on the optimal torque expression derived from the MPPT condition, shows the best performance. It quickly reaches and sustains a C_p value of about 0.49, which is indeed the best power coefficient calculated for this turbine in Wind Activity 1.

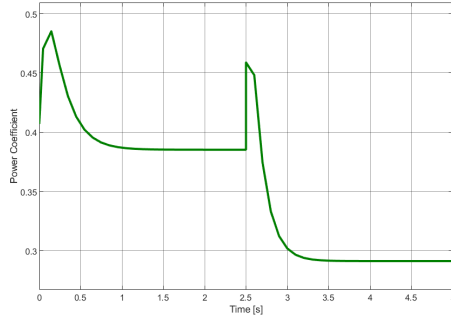


Figure 4: Power Coefficient Load 1 (Linear Torque)

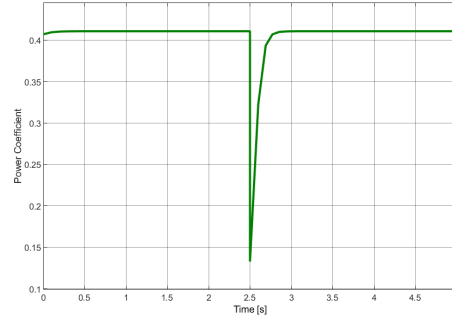


Figure 5: Power Coefficient Load 2 (Quadratic Torque)

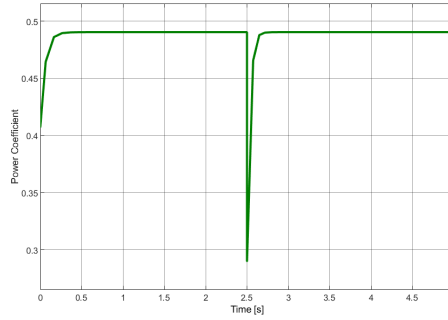
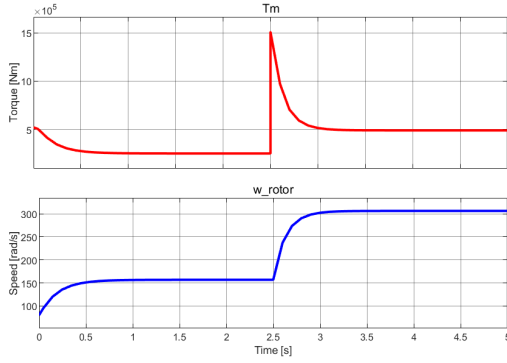
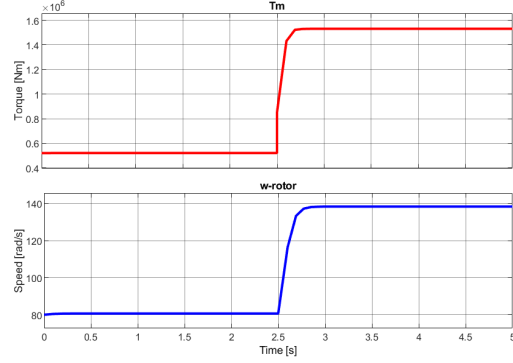
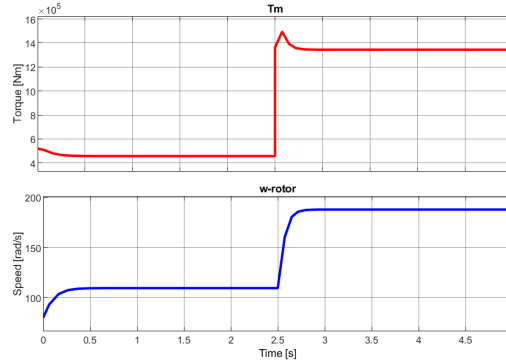


Figure 6: Power Coefficient Load 3 (Optimal Torque, MPPT)

3.2.2 Torque and Rotor Speed Plots

The mechanical torque and rotor speed results also show noticeable differences in dynamic behavior. The rotor speed increases significantly in each case, as expected under a wind speed increase. Load 1 (Figure 7) applies torque as a linear function of rotor speed, meaning the steady-state applied torque increases much less than its quadratic counterparts (Figures 8 and 9). The mechanical torque and power coefficient spike because a step change in wind speed is similarly dramatic. The power from the wind causes the torque spike since the two are directly related by the rotor speed, which does not change until after the torque changes. The high torque increases the rotor speed, and as the rotor speed updates the power coefficient and mechanical torque both settle to steady-state. The mechanical torque for the quadratic cases do not decrease nearly as much after the spike because the load torque is much higher, and the torque values are equal at steady state.

Figure 7: Torque and Rotor Speed
Load 1 (Linear Torque)Figure 8: Torque and Rotor Speed
Load 2(Quadratic Torque)Figure 9: Torque and Rotor Speed 3
(Optimal Torque, MPPT)

3.3 Simulation Results

Among the different loads, Load 3 provides the best wind power extraction. Load 2 shows that a load with a constant quadratic relation to rotor speed will have a constant power coefficient, and Load 3 uses the characteristic expressions of the turbine to pinpoint the quadratic term corresponding to the maximum C_p . Load 2 performs reasonably well as an approximation, while Load 1 is purely proportional and does not account for aerodynamic conditions, resulting in the lowest efficiency.

4 PMSG Operation under VSC Control (Task 3.2)

In the second part of the report, the behavior of a Permanent Magnet Synchronous Generator controlled through a back-to-back Voltage Source Converter is analyzed. The model used in this section is a predefined system that was provided, already including the PMSG and both machine-side and grid-side converters. The objective is to describe and understand the system from the operating scenarios.

4.1 Model Setup

In the initial model setup, the PMSG operates as a constant speed motor drawing power from the grid while spinning a torque load. The machine side VSC uses a dq-reference frame control structure to independently manage the torque and reactive power by adjusting the stator current components. The behavior of the generator is controlled with the torque reference provided to the machine VSC controller. The grid side converter maintains the DC-link voltage and controls reactive power exchange with the grid.

The examined simulations include two test scenarios. One simulation with fixed references for torque and reactive power and the second one which introduces a step change in both values at different times. Throughout both cases, mechanical torque, rotor speed and power flow at the machine and grid side are recorded and analyzed. Note that the sign convention for the simulation is that positive power flows to the grid from the grid VSC and to the PMSG from the machine VSC. The aim is to evaluate the response and controllability of the provided PMSG-VSC system before integrating it into the full wind turbine configuration in the next task.

4.2 Simulation Analysis

4.2.1 Scenario 1

In the first scenario, the PMSG is to consume 5 MW of power with no reactive power provided from the VSC. In order to determine the torque reference to provide the controller, we divide 5 MW by the synchronous speed of the PMSG (4). The current control loop in the machine side VSC includes a term for the magnetic losses, so the calculated torque reference will result in 5 MW used by the load.

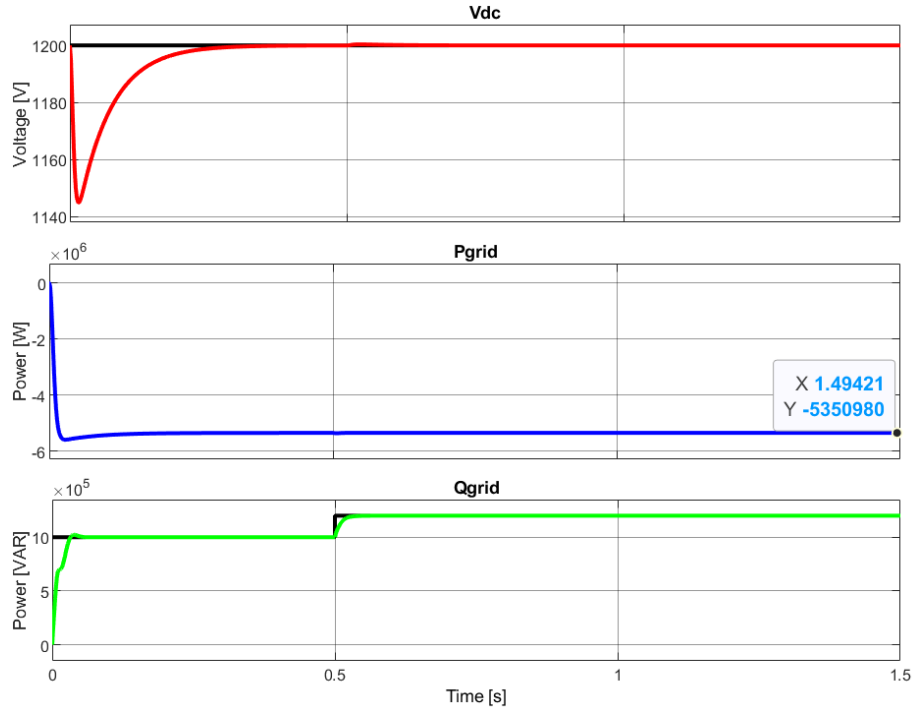


Figure 10: DC-Link Voltage and Grid-Side Active/Reactive Power Response

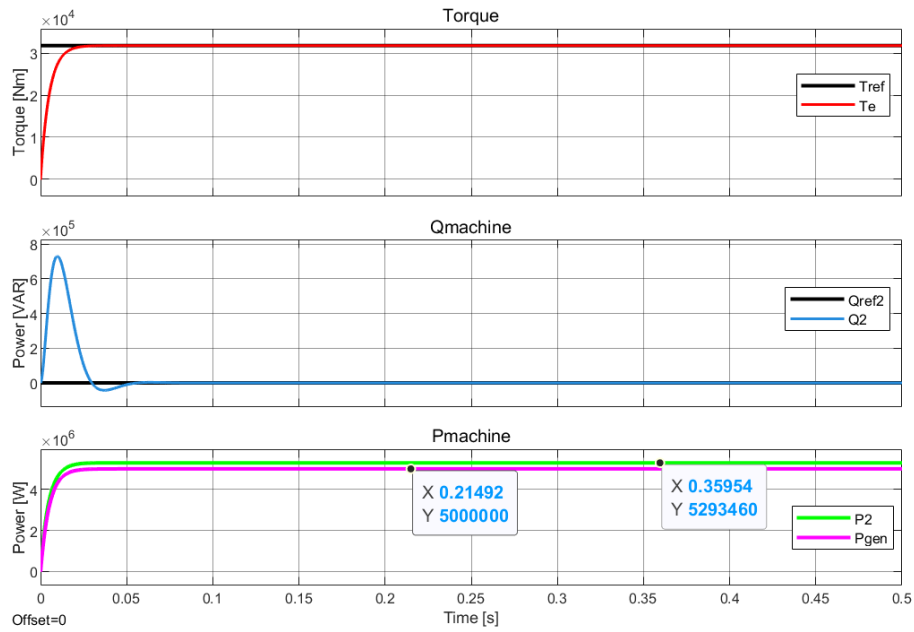


Figure 11: PMSG Torque & Reference and Machine-Side Active & Reactive Power Response

Pgen (MW)	PAC (MW)	Pgrid (MW)	Total Efficiency (%)
5	5.29	5.35	93.4

Table 2: Scenario 1 Power Transfer

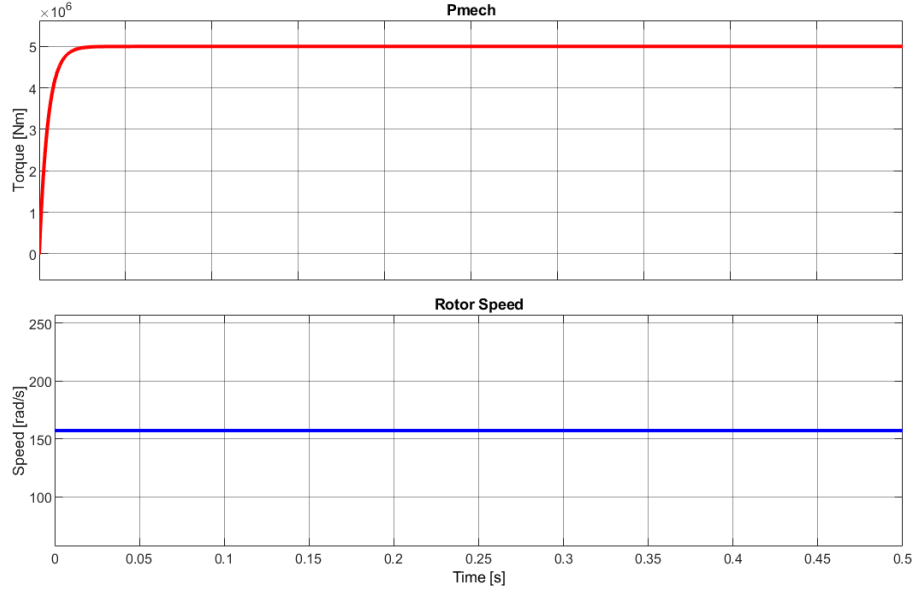


Figure 12: Mechanical Input Power and Rotor Speed

The results confirm the expected behavior: Figure 11 shows that the power consumed by the PMSG is exactly 5 MW. The amount of power supplied has to be higher at each previous stage to account for the losses, which are tabulated in Table 2. Most of the power loss in this scenario occurs between the machine VSC (labeled 'PAC') and the load. No resistive or inductive losses are modeled in this connection, so we can conclude the majority of losses occur in the PMSG. Internal SG losses are attributable to the stator core losses as well as conduction losses in the stator coils.

4.2.2 Scenario 2

In Scenario 2, a step increase is applied to both the torque and reactive power references of the grid-side VSC at 1 and 0.5 seconds, respectively.

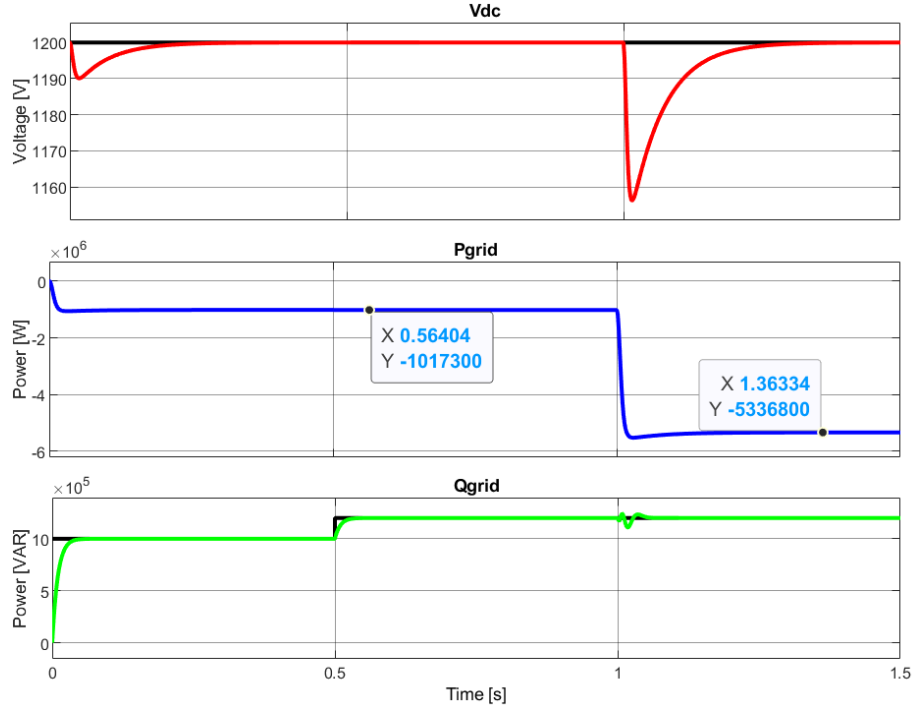


Figure 13: DC-Link Voltage and Grid-Side Active/Reactive Power Response

Figure 13 displays the DC-link voltage and the active and reactive power exchanged at the grid side. The DC voltage remains nearly constant at the nominal value until 1 second, where a significant dip occurs as expected due to the rush of output current on the machine side. The drop reflects the energy imbalance caused by the temporary delay in power transfer adjustment between the converters. The imbalance is covered by the DC link capacitor, which creates the voltage drop. The active power at the grid side (P_{grid}) shows a step decrease, corresponding to the rise in the torque reference, confirming that more active power is being delivered by the grid, since positive power flows towards it. Simultaneously, the reactive power (Q_{grid}) undergoes a small transient and stabilizes shortly after the step, indicating the effectiveness of both of the grid side VSC's control objectives.

Pgen (MW)	PAC (MW)	Pgrid (MW)	Total Efficiency (%)
1	1.01	1.02	98.3
5	5.28	5.34	93.7

Table 3: Scenario 2 Power Transfer

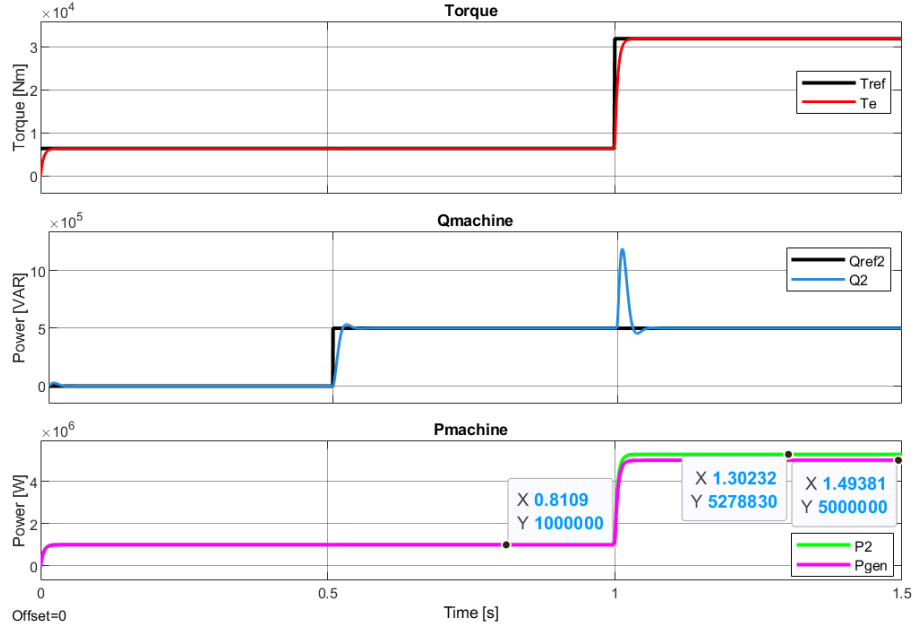


Figure 14: PMSG Torque & Reference and Machine-Side Active & Reactive Power Response

Figure 14 illustrates the behavior of the machine side converter, showing mechanical torque, machine side active power and reactive power. The active power at the machine side shows the PMSG consuming exactly 1 and 5 MW, but the power output of the VSC becomes visible after the step. The indication from the graph is confirmed in Table 3 that the system becomes less and less efficient at higher power flows. The reactive power step at 0.5s shows a small overshoot, as well as a huge reaction to the increase in torque. The torque step causes the current to increase, which necessarily means more reactive power is being delivered, since the torque current loop must increase the current magnitude to create the reactive power error. The Q controller also shows a slightly overshoot response when adjusting the output to maintain 0.5 MVAR, indicating that the speed of the dynamics of the two loops are not sufficiently separated to exhibit a first-order response.

Figure 15 shows the constant speed of the generator and that the torque references were calculated correctly to create a 1 MW to 5 MW step in the load power.

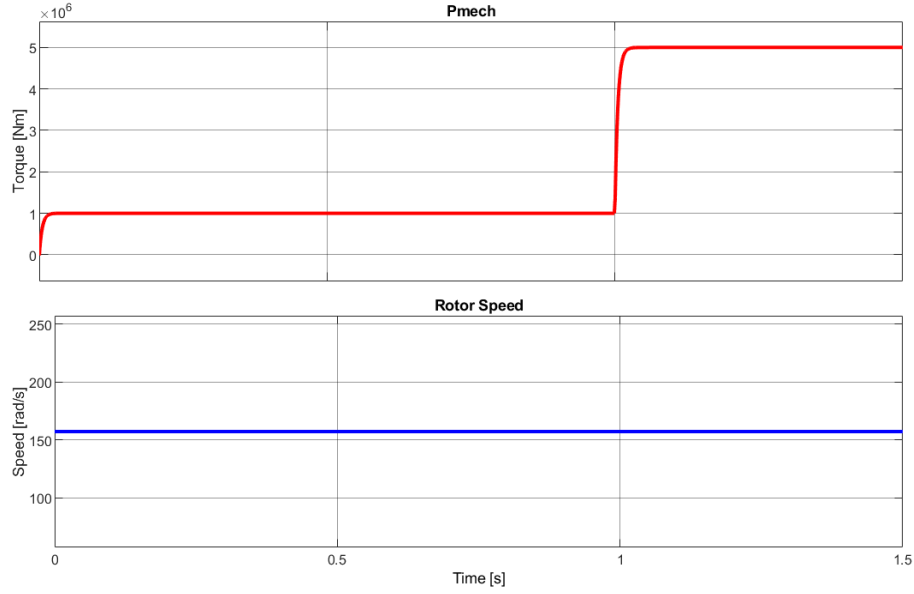


Figure 15: Mechanical Input Power and Rotor Speed

The power loss in the second scenario is largely analogous to the first. The power lost in the generator is much higher than that lost in the VSCs at 5 MW, but both parts of the system lose about the same amount of power in the 1 MW case, indicating that the hysteresis and eddy current losses in the stator core scale more significantly than the switching and conduction losses of the power electronics. Additionally, we notice that the 5 MW power delivery has a different efficiency between the two scenarios, with Scenario 2 having a higher efficiency. This is because the added reactive power increases the operating voltage of the system and decreases the necessary current to supply the set power to the machine, thereby reducing the losses. To show this phenomenon, the simulation is repeated with the reactive power step performed after the torque step in Figures 16 and 17.

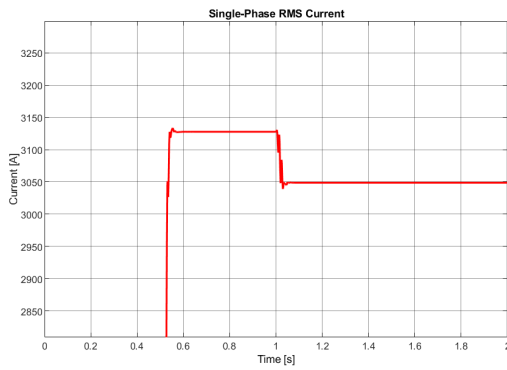


Figure 16: Machine-Side RMS Current

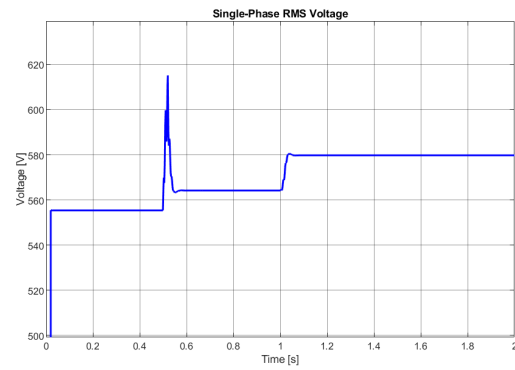


Figure 17: Machine-Side RMS Voltage

5 Complete the Type 4 model (Task 3.3)

Lastly, the previous two models are integrated together so that the machine-side VSC torque reference is now set directly to the optimal load based on the simulated wind speed as in Equation 5. The sign of the reference torque is made negative to simulate generation instead of motor operation. To test the integrated system, the wind speed input to the Tref calculation will be stepped from 5 to 8 m/s at 0.5 seconds and from 8 to 14 m/s at 1 second. The results of the simulation are shown below.

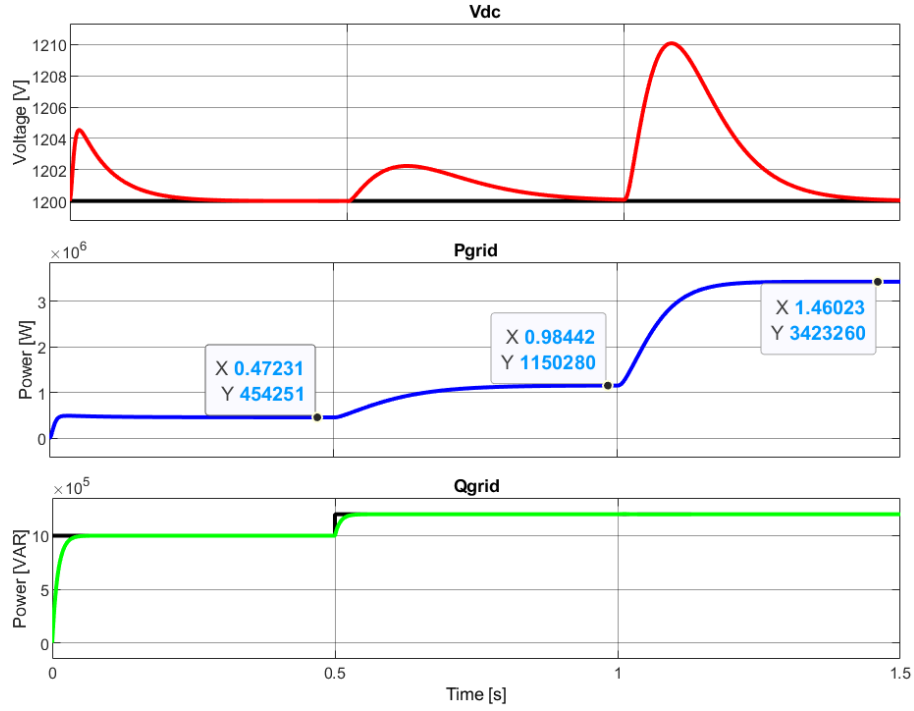


Figure 18: DC-Link Voltage and Grid-Side Active/Reactive Power Response

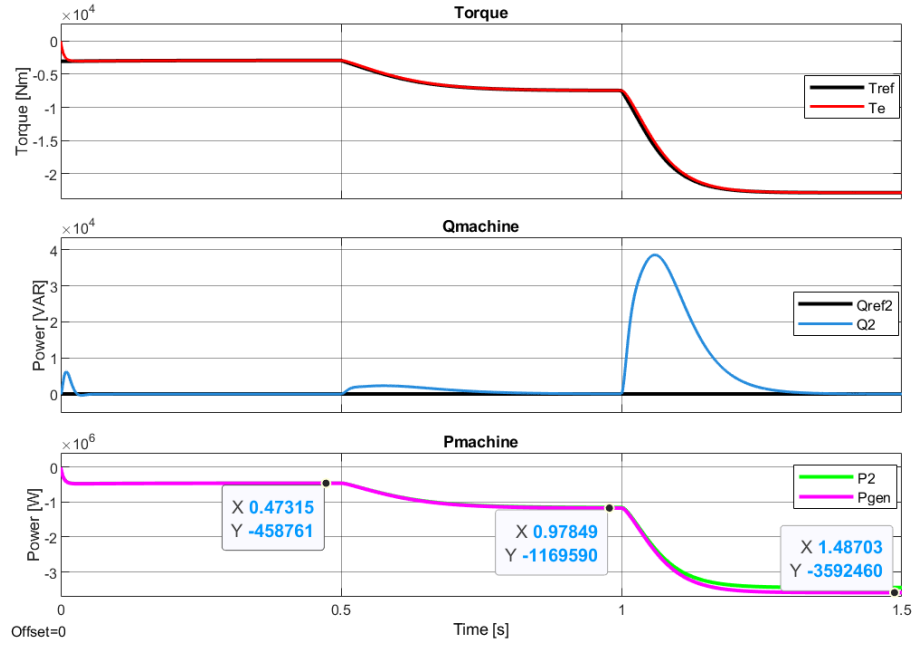


Figure 19: PMSG Torque & Reference and Machine-Side Active & Reactive Power Response

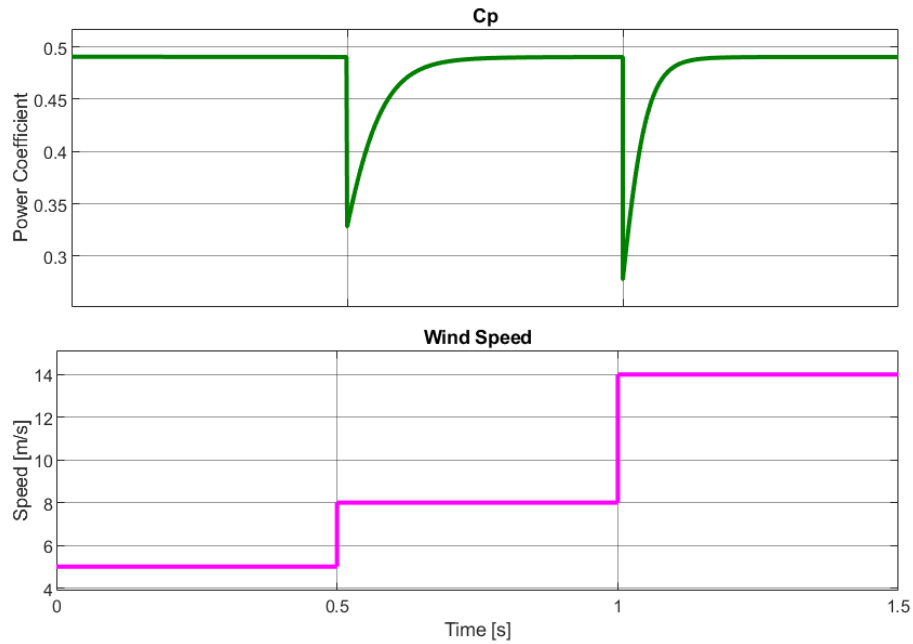


Figure 20: Power Coefficient and Wind Speed

First, the shapes of the power graphs have changed as expected. The increasing wind speed raises the torque and power delivered to the grid as more power is captured by the rotor. Since the sign convention of the power is positive towards the grid and towards the synchronous machine, in generation mode the power would be negative on the machine side and positive on the grid side, which matches the results.

Wind Speed (m/s)	Pgen (MW)	PAC (MW)	Pgrid (MW)	Efficiency (%)
5	0.46	0.46	0.45	99
8	1.17	1.15	1.15	98.3
14	3.59	3.45	3.42	95.3

Table 4: Combined Model Power Transfer

The two increases in wind speed have similar but inverse results to the previous section because of the negative torque reference. The DC voltage increases transiently rather than decreasing because of the increase in current flowing from the generator to the DC link capacitor rather than the inverse. The power and voltage increases are also much larger for the second step, since there is a larger wind increase and the power available in the wind is cubic.

The power flow results in show the efficiency of the combined system also decreases at increased power, as we would predict from the previous section. Additionally, the same phenomenon of the increasing proportion of generator losses is present in these results.

6 Conclusion

This report has examined the control and operational behavior of Type 4 wind turbines, which use back-to-back VSCs between the turbine's Permanent Magnet Synchronous Generator and the grid. The first section examined the response of a turbine to various torque loads on the generator using the characteristic equation for power coefficient. The second section observed the typical control scheme of a Type 4 turbine, with VSCs using their degrees of control freedom to set a reference load torque for the PMSG, to maintain the DC-link voltage, and to maintain reactive power sharing along the whole connection to the grid. Lastly, the first model was combined with the first to generate the torque reference for the machine VSC based on simulated wind speed.